

LSGI3320 Building Information Modelling & 3D GIS

Group Report

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Students

[Redacted Student Names]

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Abstract

This report provides a comprehensive analysis of Building Information Modeling (BIM), highlighting essential aspects such as work distribution, the development of a final 3D model, and various visual representations including solar studies, camera views, and walkthroughs. It defines key BIM terms and explores five main features of BIM. The report also outlines the four major roles of BIM in project delivery, discusses the concept of Level of Detail (LOD), and explains dimensions in BIM alongside the Information Foundation Class (IFC). Furthermore, it details the process of developing both 2D and 3D BIM models and addresses the difficulties encountered during implementation, offering insights into the challenges faced by professionals in the field.

Background

BIM is an intelligent 3D model that plays a crucial role in managing, coordinating, and simulating the entire lifecycle of buildings and construction projects. This includes phases such as planning, design, analysis, documentation, construction, renovation, and demolition. The application of BIM spans various sectors, particularly within Architectural, Engineering, and Construction (AEC) projects, where it enhances efficiency and accuracy.

BIM provides a unified platform for creating and managing 3D models, facilitating collaboration among a diverse range of stakeholders. These include AEC professionals, suppliers, clients, and CEOs. By integrating various disciplines into a single model, BIM facilitates improved communication, increases efficiency, ultimately leading to better project outcomes.

Creating a BIM model allows team members to develop a variety of essential skills. As the team builds a 3D model together, one might learn the communication skills and software knowledge in BIM. The hands-on experience and collaborative effort serves as a practical exercise that mirrors real-world BIM applications, providing insights into the challenges and opportunities present in actual BIM projects. By engaging in this process, we not only did enhance our technical skills but also have deveolped a deeper understanding of how BIM can transform project delivery and management in the construction industry.

Objective of our BIM Project

We aim to gain hands-on experience with BIM by developing a 3D model collaboratively, and apply the communication skills by resolving the problems as we progress. Furthermore, we aim to gain insights of how our BIM project reflects the real world applications of BIM. Last but not least, we will be explaining the key terms of BIM based on our knowledge from the lessons and how it can connect the project.

Workflow Distribution

Group members:

Student 1: [REDACTED]

Student 2: [REDACTED]

Student 3: [REDACTED]

Student 4: [REDACTED]

Student 5: [REDACTED]

Student 6: [REDACTED]

Student 7: [REDACTED]

Workflow assignment

This group consists of 7 members and there are 8 portions. The portions are divided equally among the members, with one of the member drawing an extra portion. For LOD4 features like furniture, they are added depending on the BIM skill of the member. Below shows the workflow assigned or taken by each member.

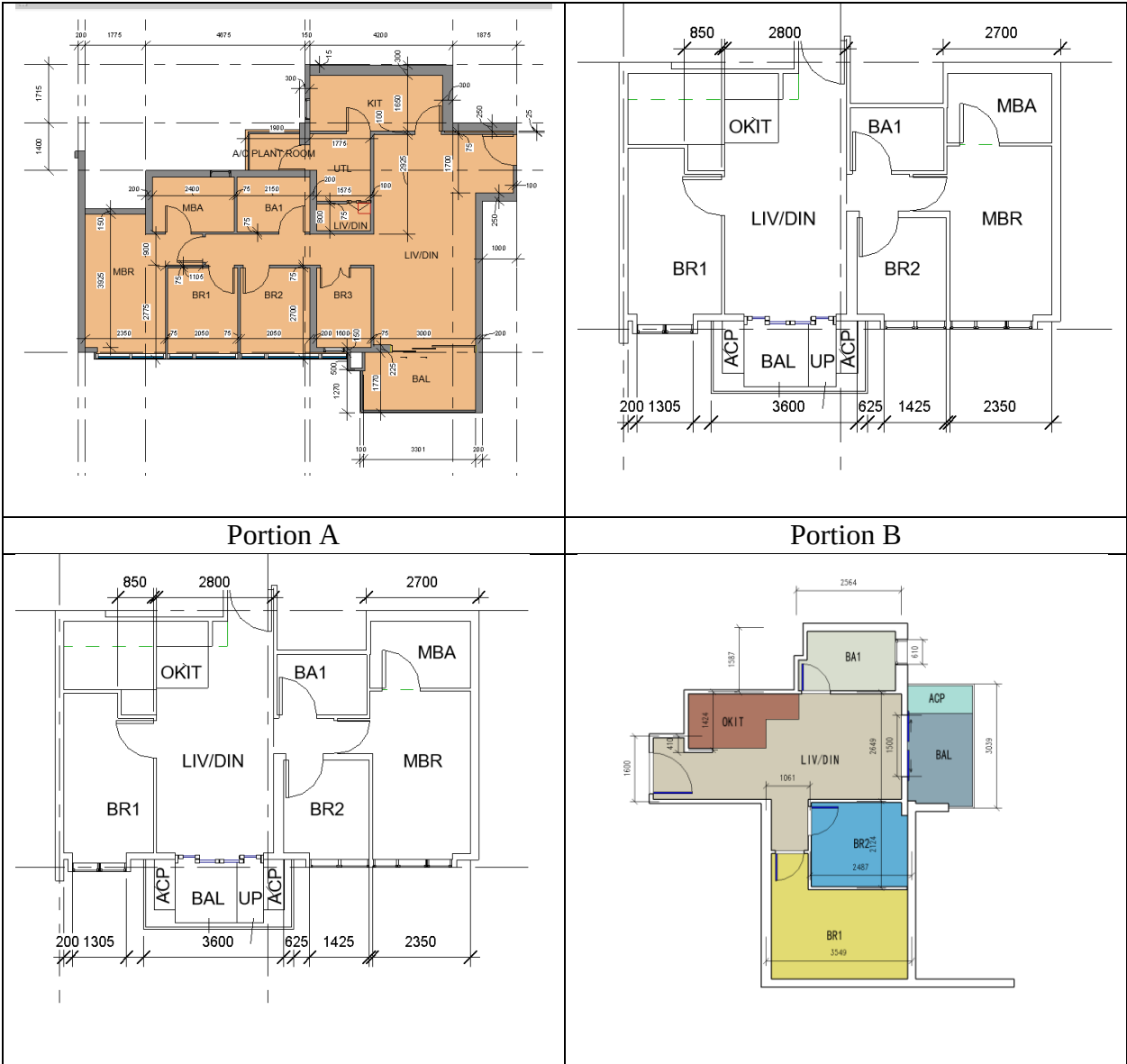
Assignment \ Student	1	2	3	4	5	6	7
Portion A							
Portion B							
Portion C							
Portion D							
Portion E							
Portion F							
Portion G							
Portion H							
External wall and window							
Internal – doors and furniture							
Solar study and walk through view							
Camera View							
Floor							
Combining buildings							
Annotations							

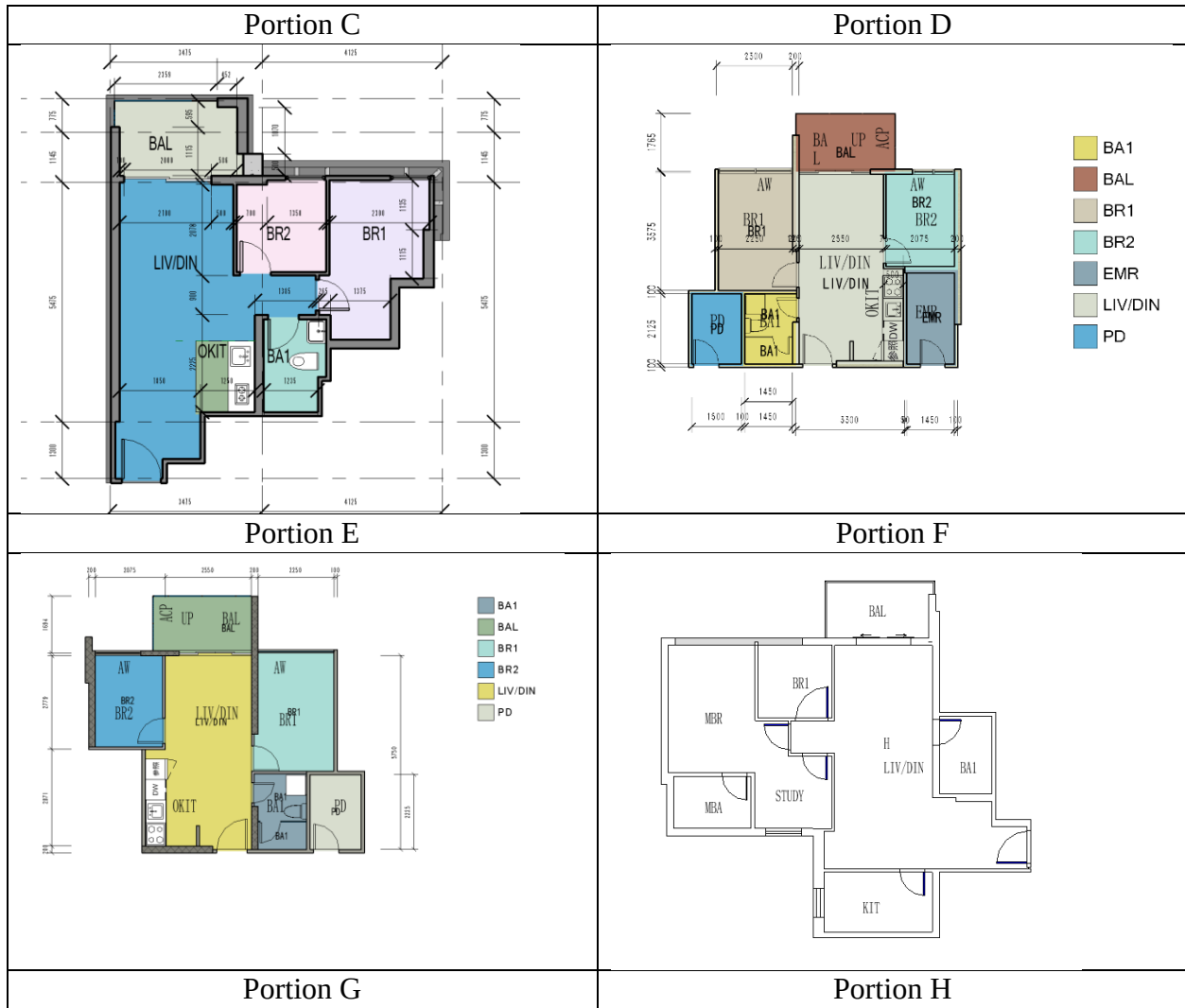
Adjustments & Detailing							
Report							

Review - Intermediate Stage

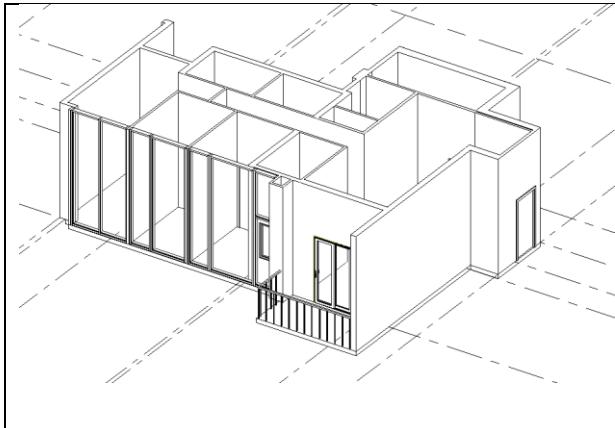
We have created floor plans and 3D views of the BIM models in the previous stage. This part serves as a review.

Drafted Floor Plan

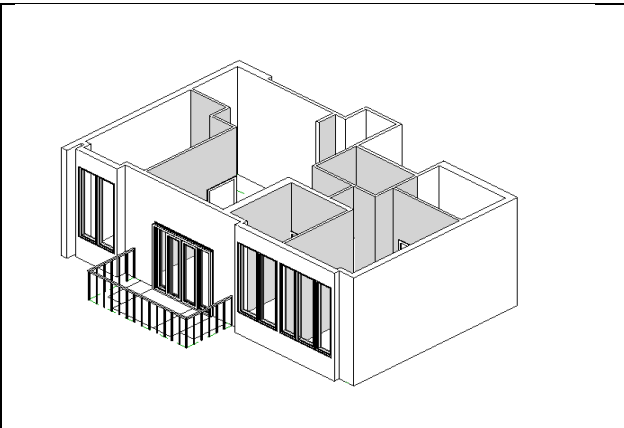




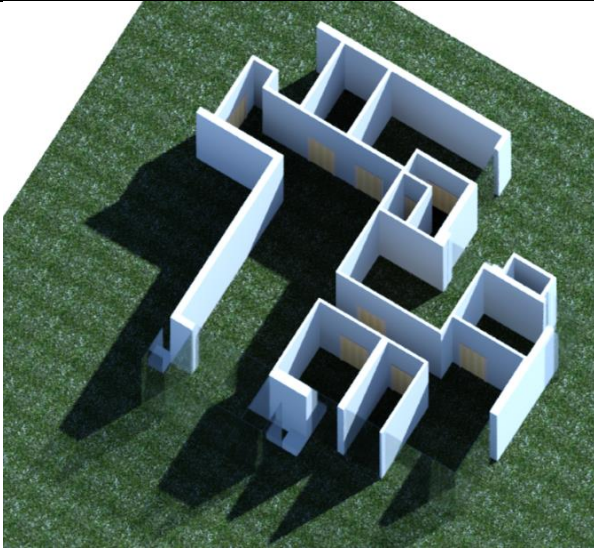
Drafted 3D view



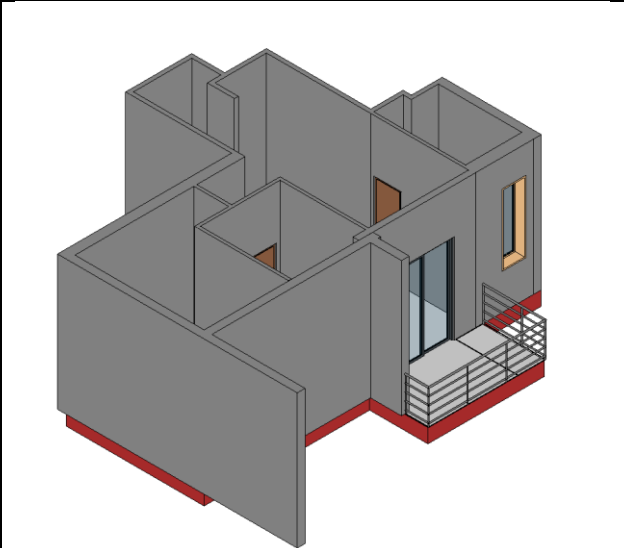
Portion A



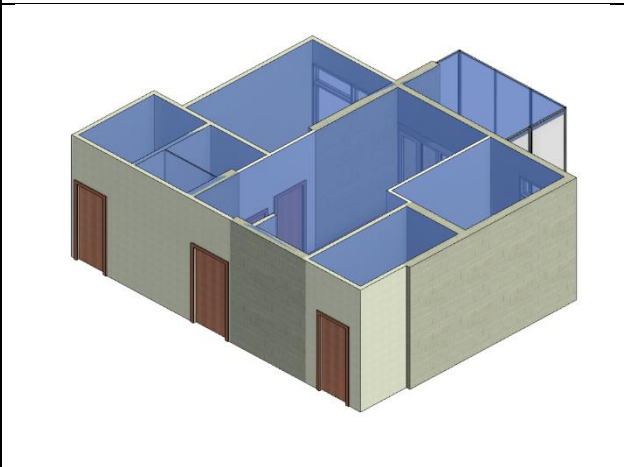
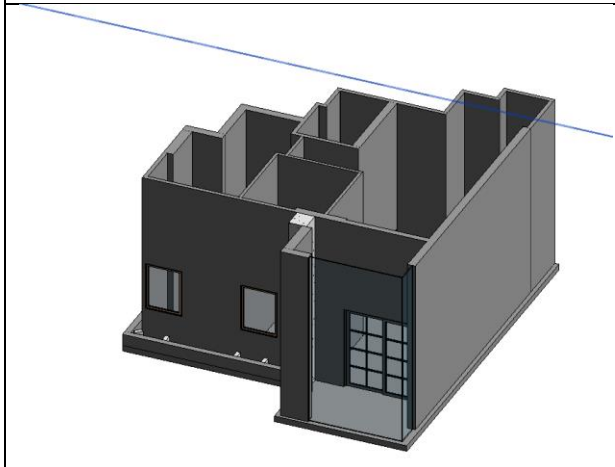
Portion B

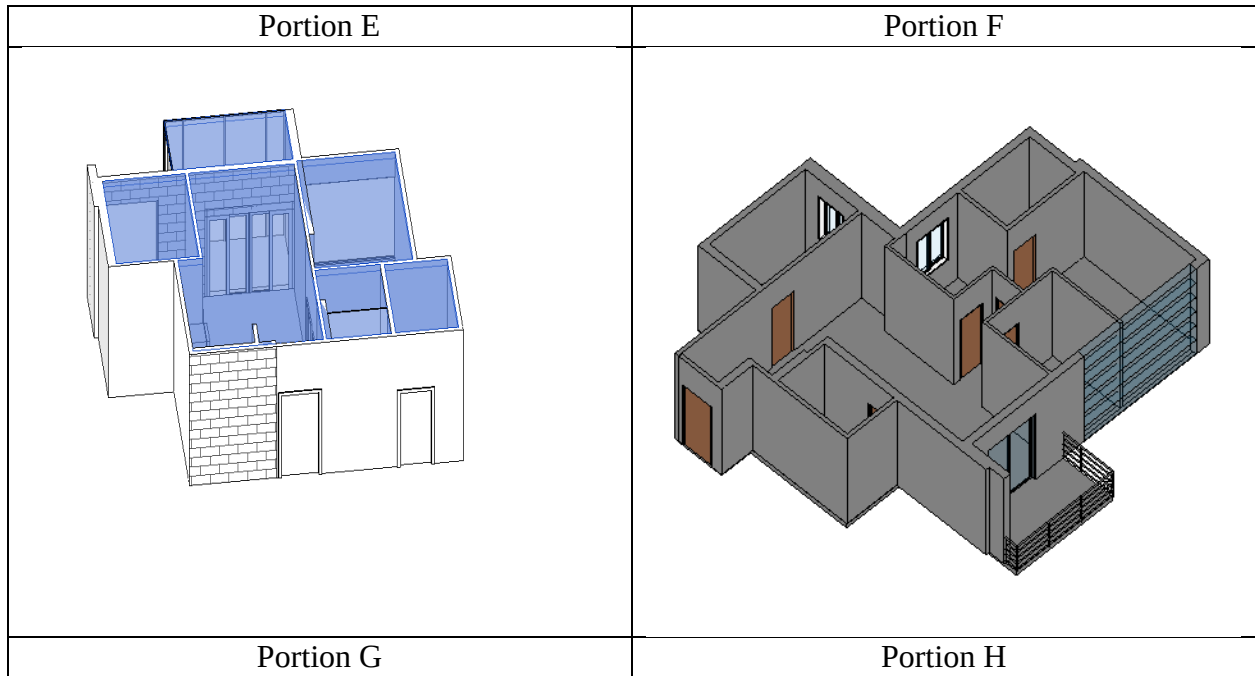


Portion C



Portion D





Reference drawing



Project Progress

We have adhered to the following steps in the development of our 3D BIM drawings using Revit 2025:

Firstly, our group created various plans (Portion A to H), ensuring that the relative levels of the floor plans were consistent from Level 0 to Level 1. It is crucial to avoid any offsets or unconstrained levels, as these can lead to height discrepancies later when combining the buildings.

Secondly, we drew all wall structures in 2D, using the provided floor plan as a reference. It is crucial in this step to ensure the walls have the correct width. As for the material and texture of the walls, the wall type can be easily modified later. We have also placed windows and doors in appropriate locations, Revit provides the function Load Family from the Insert tab to import the doors and windows, while furniture was imported from Content Packs. Additionally, we employed the Annotate feature to label room types and dimensions clearly.

Thirdly, we utilized the Link Revit function to import 3D models into the main project. It is worth noting that Revit is not forward compatible and cannot open models created in newer versions. Therefore, it is important to use the latest version of Revit. We made adjustments to the floor and wall types to ensure consistency across the features, enhancing the overall coherence of the BIM model. Thanks to the strict adherence to the relative level, we were able to merge the buildings with the same level and quickly make adjustments. For example, we lowered the height of the buildings from 4m to 2.5m by simply changing the height of the reference level (Level 1).

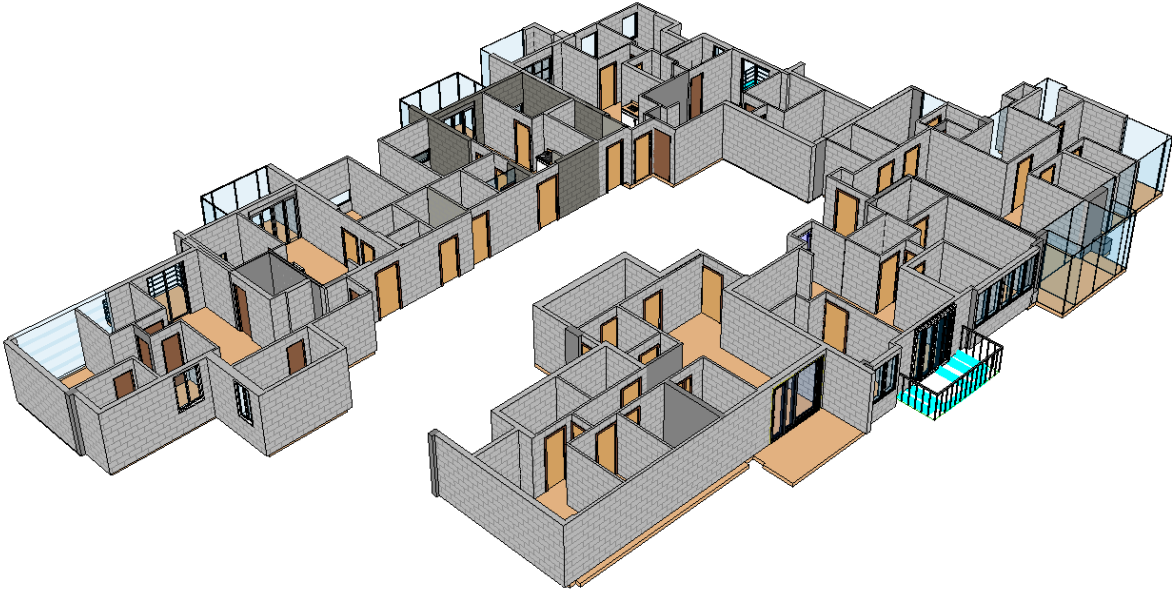
Final Product

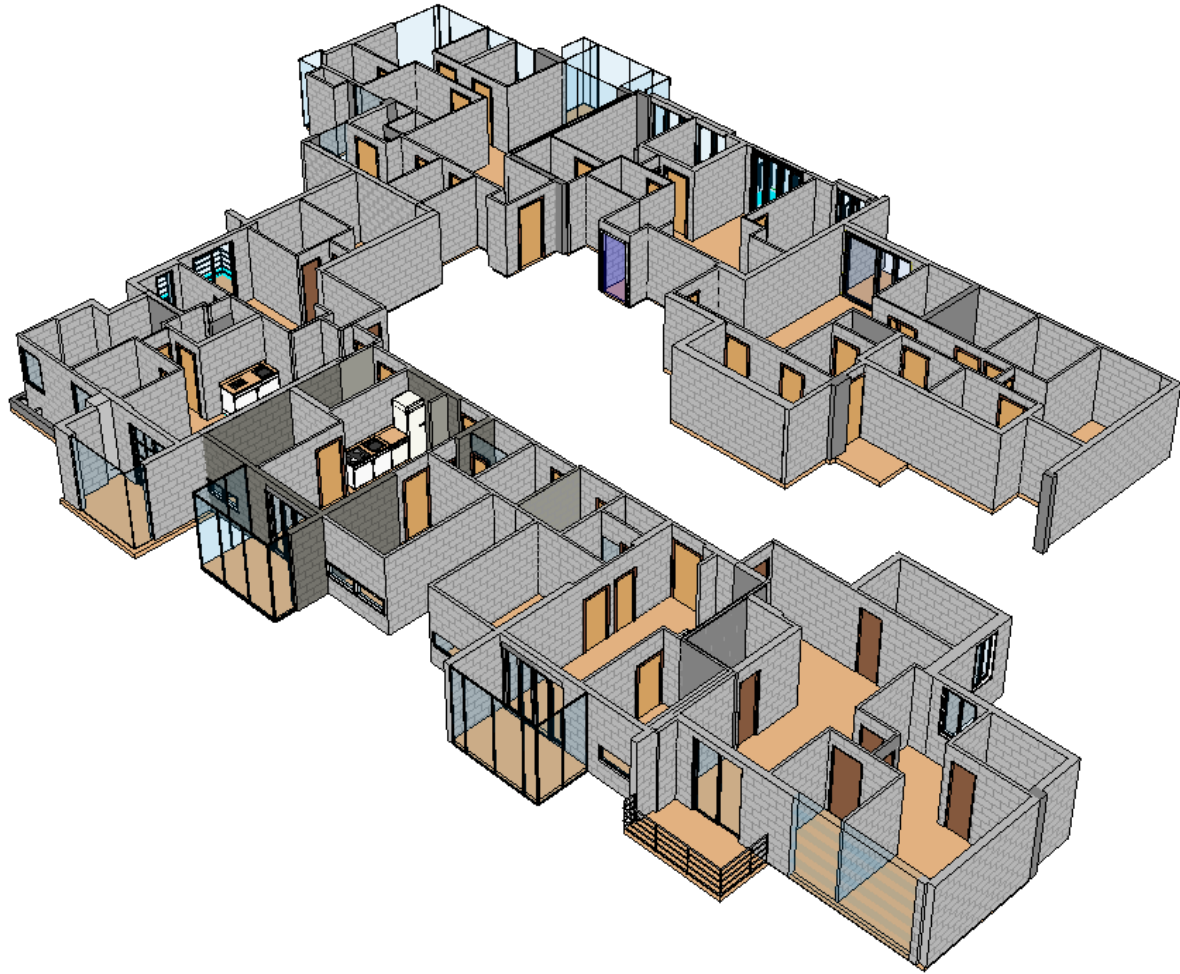
Integration of the intermediate 3D models

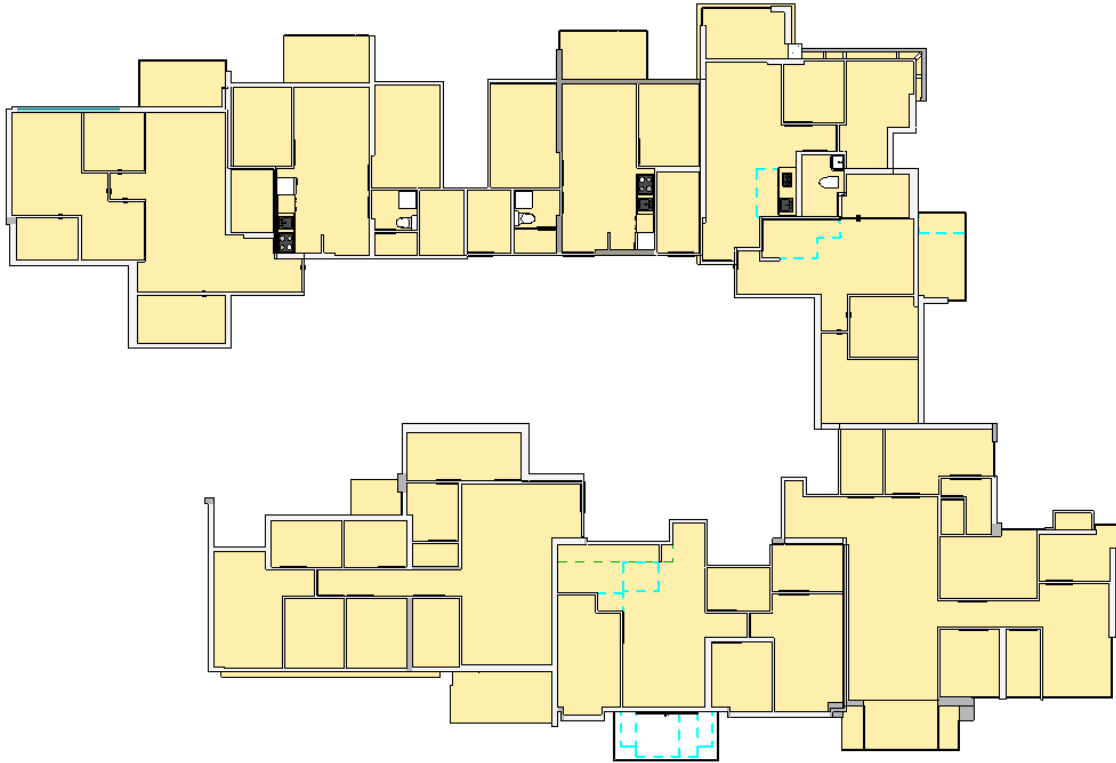
Despite the challenges in the previous stage, where we were unable to link every model, we successfully gathered the 3D models in the final stage. We have linked the 3D models using the function Link Revit. We have also unified some of the materials used in the BIM model. For example, the wall used is concrete and the texture is brick, in addition, the class name of the walls and floors are unified. As for the level of the building, they are from Level 0 to Level 1 with 2.5m height. Since Revit uses a local projection coordinate system (PCS), with the height set to the same starting and ending names, we can ensure that the combined buildings have the same height without further adjustments.

However, we still have some discrepancies in our design, for example, for the balcony of the buildings, many uses curtain walls while some uses fencing, one reason could be due to the floor plan of the building does not specify how the balcony should be enclosed. This could be one of the challenges in actual BIM projects, as the modelers might have different interpretations of the

plan and might be different from what the client desires. Therefore, effective visualization and collaboration is needed to ensure that everyone is satisfied with the final product. Since the drawing does not specify the types, and this model's purpose is for us to learn the modeling process of BIM, we could leave the discrepancies for now. It is more important to note that these problems might arise, and we should be aware that different people might have different opinions on the same design.





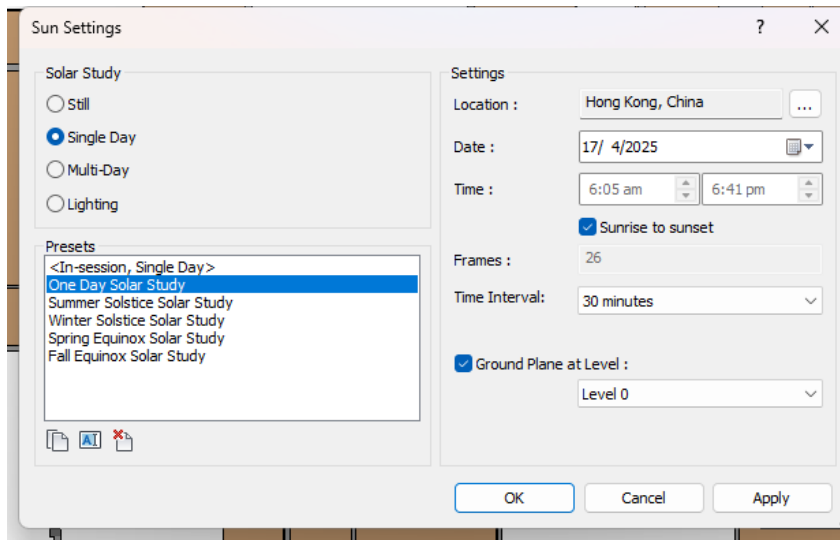


Solar Study View

The Solar Study View in a BIM is a visualization method used to analyze the impact of sunlight on a building or structure. The view tracks the sun's path and shows the areas with sunlight and shadow at different times of the day or year. In the real world, this view is likely useful for architects and designers to optimize the placement of the windows and the building's orientation, the lighting required inside the building to compensate for decreased natural lighting.

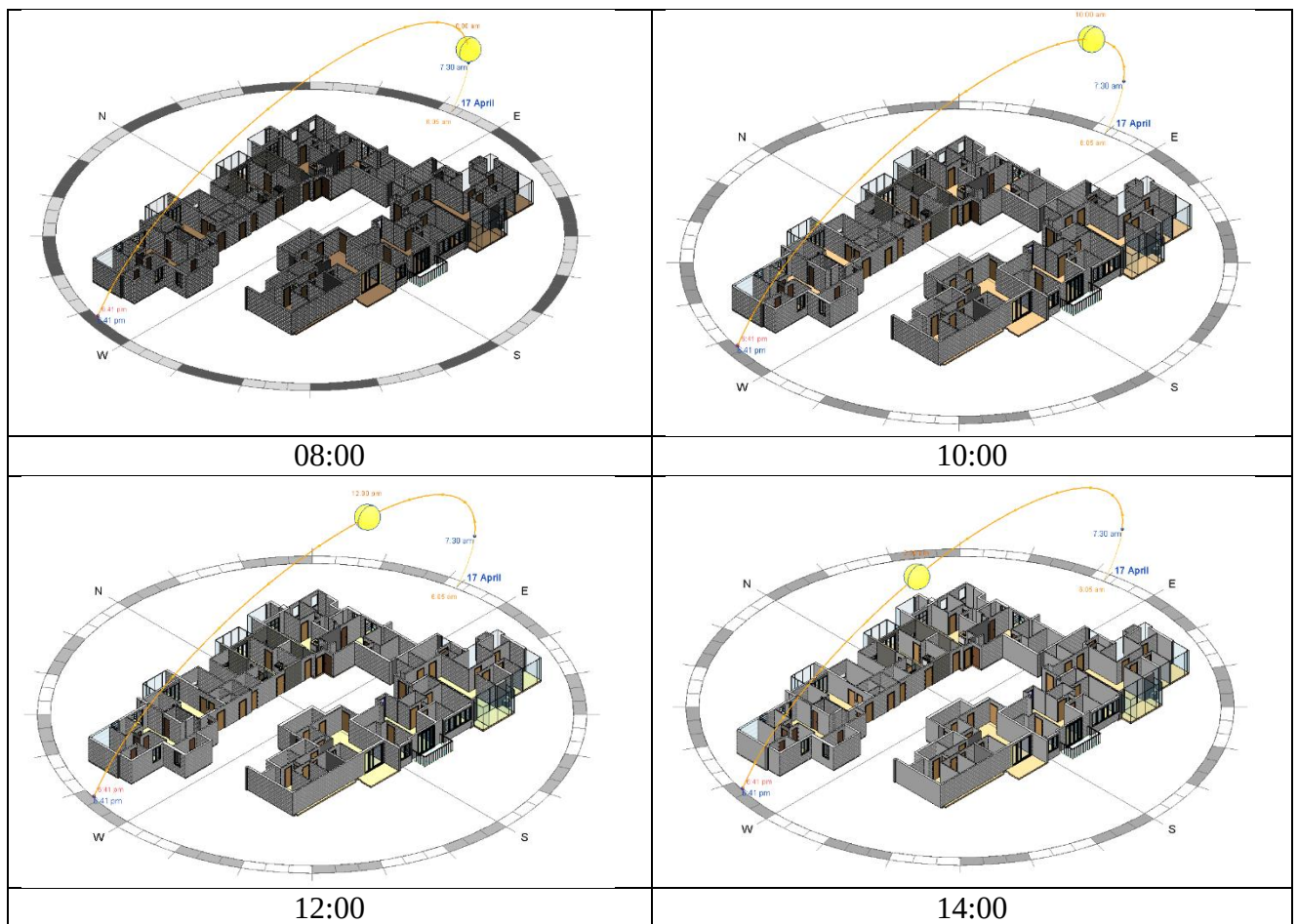
Furthermore, the buildings might accumulate more heat from the solar radiation, and with the help of solar study view, this might improve the results for the thermal analysis of the buildings. Overall, the solar study view might enhance the efficiency of the building and the overall comfort for the people in the building.

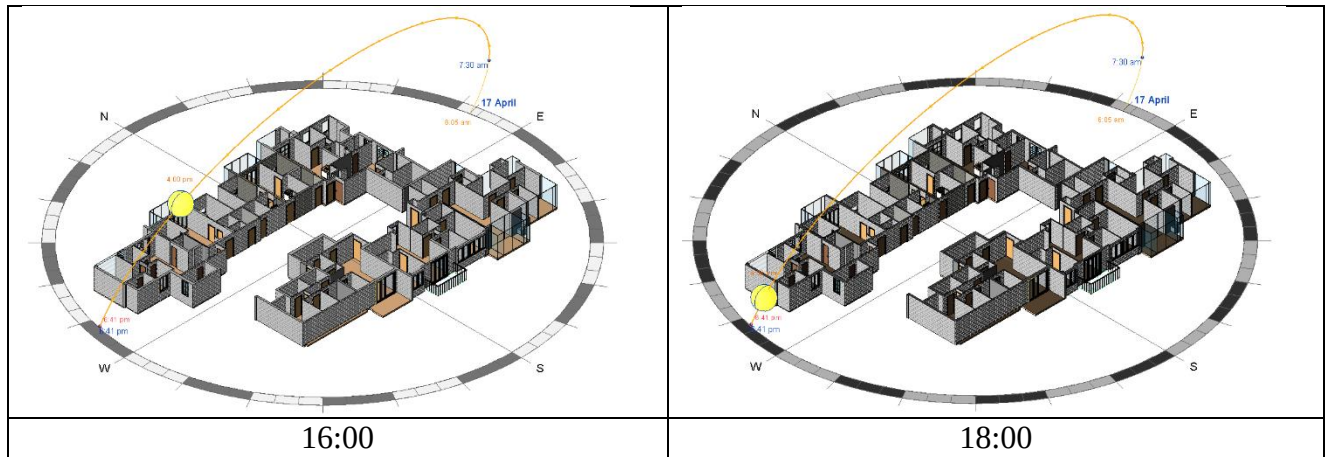
In this project, we apply the solar study view by turning on the sun path and configuring the sun settings. For example, we have selected One Day Solar Study and set the location to Hong Kong. For the date and time, we have set it to 17/4/2024, and spanning from sunrise to sunset. For the time interval, we capture the screenshots of the view every 2 hours. We think that it is a suitable interval for balancing subtle and gradual changes between the views.



Solar Study settings

Solar Study View Results



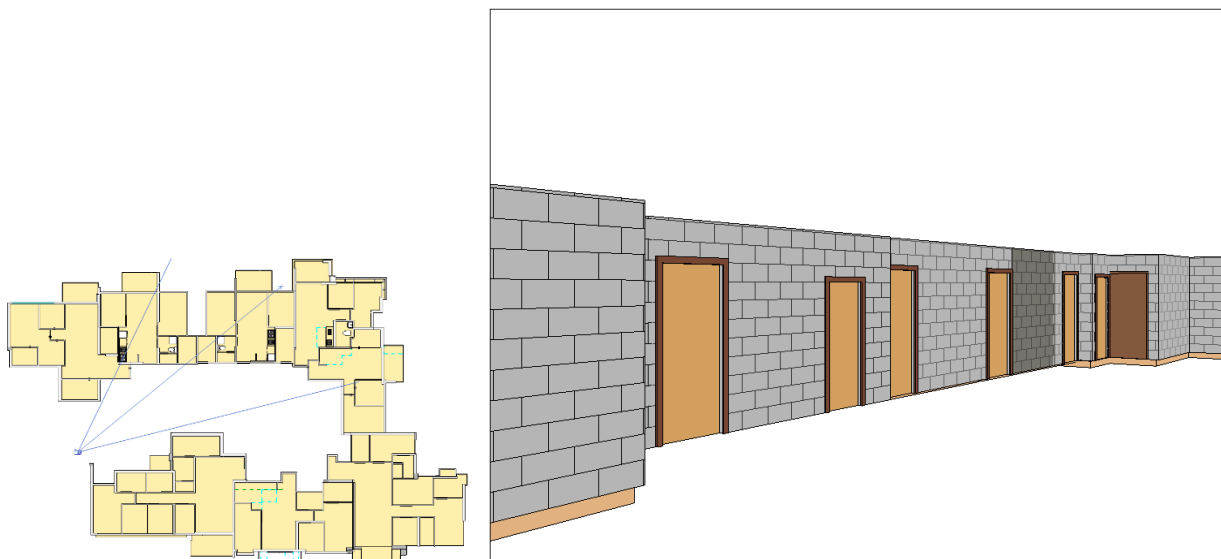


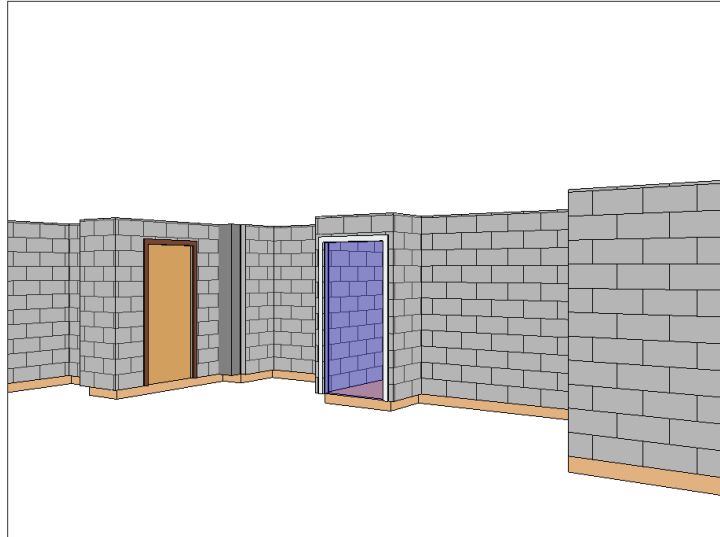
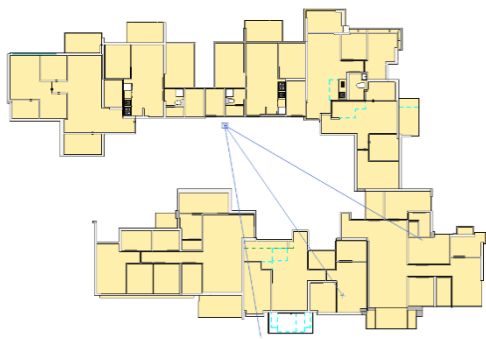
Camera View

Camera view is a feature in Revit used to capture the point of view from the point of the camera viewing in a particular direction. It is also a key feature for visualization for both AEC and non-BIM users. In the real-world applications, BIM users might use this view to visualize the inter-visibility, the bright and dark areas, the placement of the structures or utilities, and for non-BIM users they might use this feature to view the aesthetics and design of the 3D model.

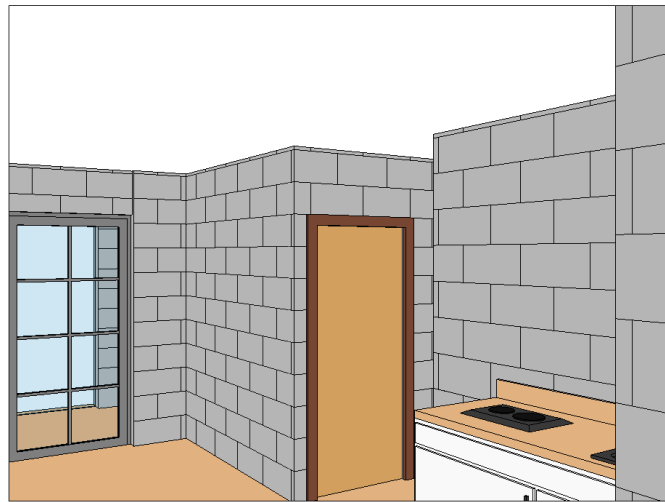
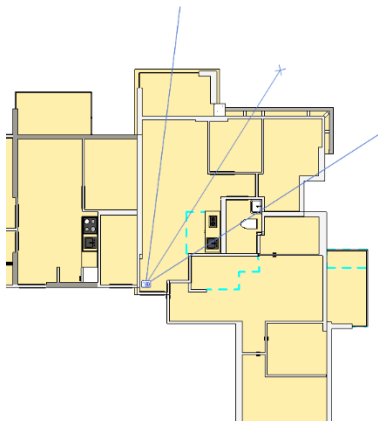
In our project, we have used the camera view to capture the interior view of the buildings, for the apartments as a whole and individual rooms.

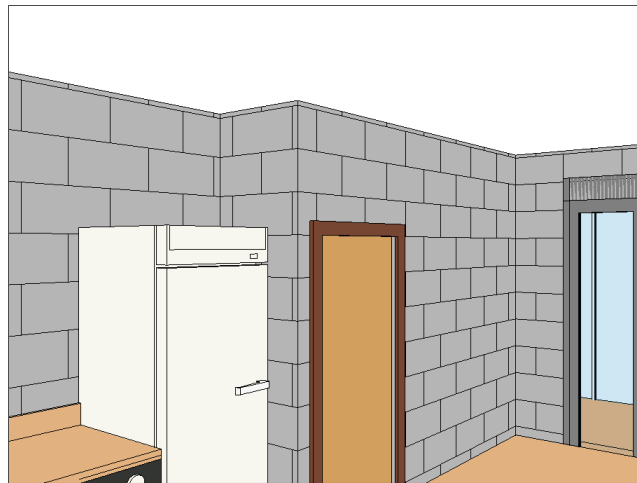
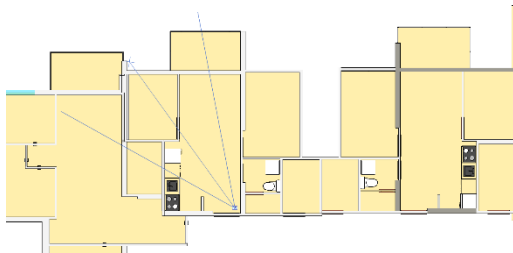
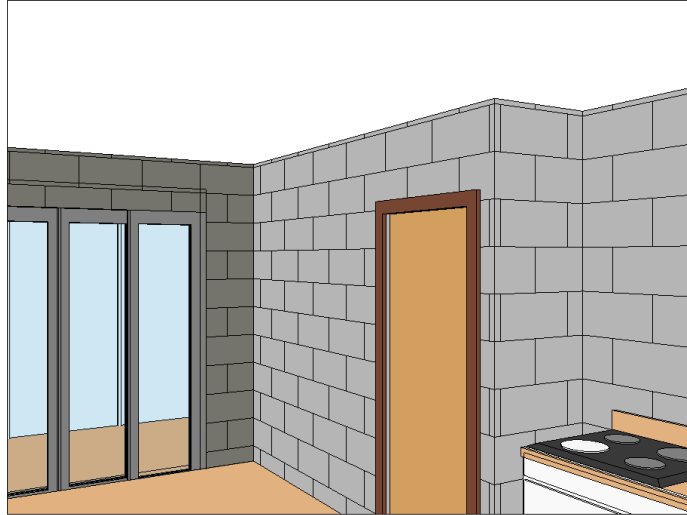
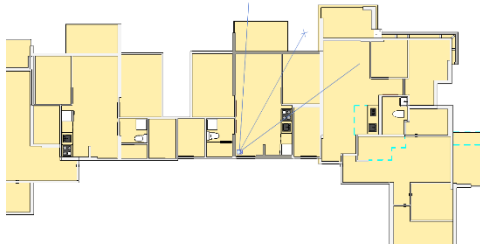
Interior of the Apartments





3.2 Interior of Rooms with furniture (Room E, F, G)





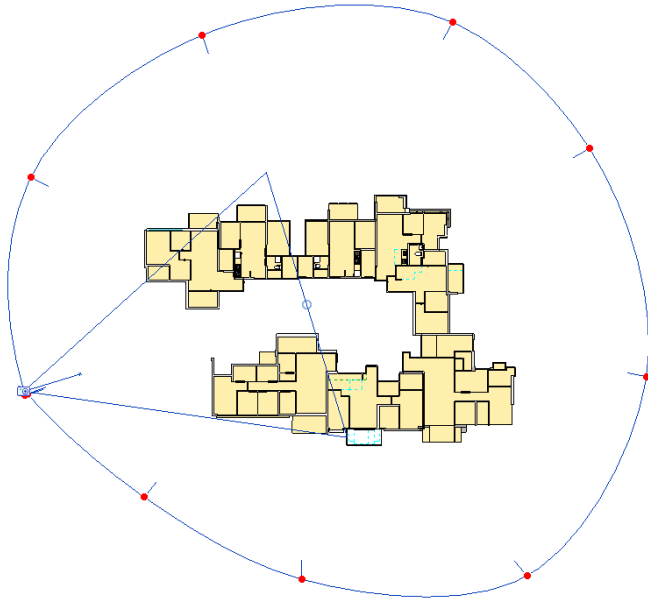
Walkthrough View

A walkthrough view in BIM provides a realistic first-person view of the 3D model, much like viewing and touring the building or structure in-person, providing a realistic and immersive experience. It is useful for visualizing the spatial layout, design, and functionality. In the real world, walkthrough view can likely be applied in visualization for non-BIM users, especially stakeholders like CEO, media, and client, it can also be used for collaboration with AEC and suppliers and the clients for better understanding of the building structure and design to make decisions for the design and construction phase.

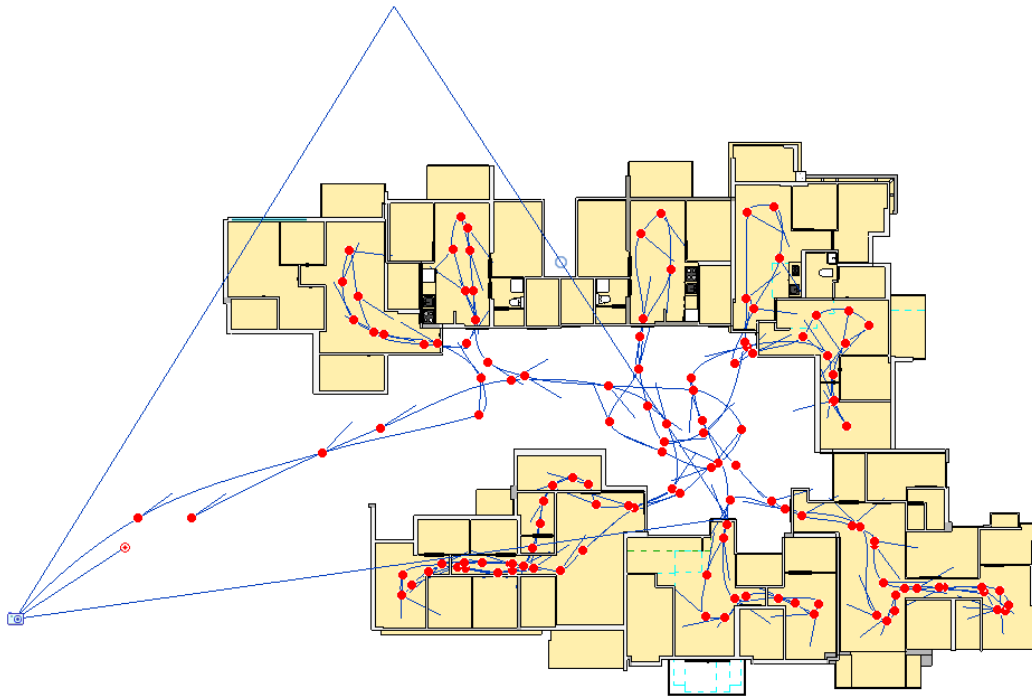
As for our walkthrough view of the BIM project, we have created two views for exterior and interior respectively. The views involve different camera positions and angles. As for the exterior view, we point the camera to the center of the building as the camera goes around it, capturing the exterior design of the building. For the interior view, we use more keyframes for more

important information, for instance the interior design for each room and even furniture in some rooms, showing quality of up to **LOD4**. To save rendering time, the colors of the view are set to “**Consistent**”, allowing the walkthrough view to be vivid without compromising the rendering frames. The videos are attached as ExteriorWalkthrough.mp4 and InteriorWalkthrough.mp4 in the Videos folder.

4.1 Exterior Walkthrough View



4.2 Interior Walkthrough View



For the video of the interior model, the starting point is outside of the building, then going between rooms A and H. The building tour starts from room H, then G, F, E, and so on finally to room A. The camera point of view also stays in the rooms for longer periods for capturing the details of the interior designs of individual rooms.

Key Terms in BIM

There are multiple key terms in BIM. The following covers the five features of BIM, BIM roles, 5 levels of detail, and multi-dimensional BIM.

Five features of BIM

There are five key features of BIM, namely: visualization, collaboration, simulation, optimization, and cartography. The following explores the five features of BIM:

Visualization

BIM transforms 2D designs into immersive 3D models, providing a realistic representation of the project. This "what you see is what you get" approach enhances understanding and communication among stakeholders. Levels of Detail (LODs) facilitate efficient visualization, allowing users to view models at varying levels of complexity depending on the project's phase. According to Kjartansdóttir et al. (2017), 3D models are often used for visualization, while 2D

models are often used for documentation. Furthermore, the high level of visualizations in BIM “enabled project team to simultaneously explore both product and process design”. BIM-based visualization interfaces are also crucial for the transparency of the process, for example construction, co-operation between stakeholders, ultimately maintaining the flow in managing the projects.

Collaboration

BIM assists communication and decision making by providing a common platform among the team members and different stakeholders such as AEC professionals and the non-BIM users, including clients, government, CEOs, and media. According to Kjartansdóttir et al. (2017), “Collaboration is the heart of the BIM process”, it greatly benefits the construction industry, often increasing the productivity, quality, and cost. Ultimately, BIM improves construction efficiency through effective communication.

Simulation

BIM allows for comprehensive simulations of the real world, for instance the fire, thermal, emergency, wind simulation of the buildings and structures. This helps the team to assess the performance of the buildings and identify the potential issues before construction begins. This feature assists decision-making of the stakeholders, reducing risks ahead of time, and enhancing project outcomes.

Optimization

Optimization is a key feature of Building Information Modeling (BIM) that helps make projects more cost-effective. The optimization process within BIM software is virtual, allowing it to be run multiple times to test different scenarios. This optimization is time-effective, as it can analyze factors like material integrity and structural stability without the need for physical prototypes. Ultimately, the optimization capabilities of BIM make it easier to plan for the long-term maintenance and upkeep of buildings and infrastructure projects.

Cartography

BIM supports cartography by converting 3D models into 2D maps, enhancing interoperability with Geographic Information Systems (GIS). This integration combines large-scale geographic data with detailed building information, utilizing georeferencing systems and coordinates used in surveying to create accurate maps derived from BIM models.

Roles of BIM

As for the roles of BIM, it can be subdivided into four major roles: modeler, engineer, project manager, director.

Modeler

The Modeler is responsible for importing maps and documents into BIM models. This role involves creating accurate digital representations of physical structures, ensuring that all relevant data is integrated effectively into the BIM system. According to BreakWithAnArchitect (2023), a BIM modeler ensures “the model aligns with the standards and goals set in the BIM execution plan and BIM project workflow criteria, creates discipline-specific BIM modeling content, and develops accurate construction drawings and extracts data based on the discipline BIM platform.

Engineer

Engineers generate BIM models and collaborate closely with designers, construction professionals, and other stakeholders. Their expertise ensures that the technical specifications are accurately represented within the model, facilitating effective design and construction processes.

Project Manager

The Project Manager coordinates technical tasks and serves as the primary point of communication between project owners and various parties involved. Responsibilities include generating the BIM application platform, creating standardized workflows, and studying new technologies to enhance project efficiency and outcomes.

Director

The Director plays a strategic role by integrating different resources and planning for various departments. This position involves summarizing and reviewing specific results, conducting state-of-the-art BIM research, and exploring the applicability of technical innovations to improve project outcomes.

Level of Detail (LOD)

For the five levels of detail (LOD) of BIM, it consists of from LOD0 to LOD4.

LOD0

LOD0 represents the most basic level of detail, where buildings are depicted as flat polygons on a flat map, or sometimes 2.5 dimensional aerial photographs. This level is often used for large-scale overviews or city planning where precise details are not required. The emphasis is on the overall footprint and location of the buildings rather than their specific shapes or features.

LOD1

At the LOD1 level, the buildings are represented as simple 3D blocks, with polygons that have height information. This provides a more realistic representation of the building's volume and

mass, allowing for better understanding of the overall scale and proportion of the structures. LOD1 is useful for applications that require a basic 3D understanding of the built environment.

LOD2

The LOD2 level introduces more detailed building shapes, including the representation of the roof structure, windows, and walls. This level of detail allows for a more accurate and realistic depiction of the building's architectural features. LOD2 is often used in applications that require a higher level of visual fidelity, such as urban planning, architectural design, and visualization.

LOD3

At the LOD3 level, the focus shifts to the external features of the buildings, including detailed representations of windows, doors, and the materials used for the walls. This level of detail provides a more immersive and realistic experience, allowing for a better understanding of the building's appearance and aesthetics. LOD3 is commonly used in applications that require a high level of visual realism, such as virtual tours or architectural visualization. According to Uusitalo et al. (2019), LOD3 can be subdivided into LOD300, LOD325, LOD350. LOD300 is the starting point for detailed model development in the LOD3 family. It is slightly more detailed than LOD290, as the whole model is fully coordinated. LOD325 is clash-free, and LOD350 contains procurement-level information such as walls, pillars, beams, and hollow core slabs.

LOD4

The LOD4 level represents the most detailed and comprehensive representation of a building, including the internal features such as furniture, doors, and utility pipes. This level of detail is often used in applications that require a complete digital representation of a building, such as facility management, building maintenance, or detailed design and planning. LOD4 is the most resource-intensive level and is typically reserved for specific use cases that warrant an additional level of detail.

Dimensions of BIM

According to NBS (2021), BIM modeling consists of different levels of context, usually referred to as dimensions. In BIM dimensions, there are 3D, 4D, 5D respectively. There are also higher dimensions in BIM, like 6D and 7D.

3D BIM

Represents spatial dimensions (x, y, z) of the model.

4D BIM

Incorporates time, allowing for scheduling and project timelines.

5D BIM

Includes cost, providing financial analysis and budgeting.

6D BIM

Focuses on facility management, enhancing operational efficiency post-construction. However higher dimensions like 6D or 7D are disputed.

7D BIM

Addresses sustainability, integrating environmental considerations into the project lifecycle.

Industry foundation class (IFC)

Industry foundation class is a global standard for BIM data exchange, and BuildingSMART is the maintainer of the IFC, with its standard is open and international under ISO 16739-1:2024. According to BuildingSMART (2025), the IFC schema is standardized data model that defines the identity and semantics, attributes, relationships of objects, abstract concepts, processes, and people. IFC is usually used to exchange BIM data between parties. For example, IFC plays a role of the neutral third-party bridging between the architect and the owner, as the architect provides the model of a new facility design and the owner might be sent an as-built model of the necessary information. IFC is under the common formats, most commonly in STEP, but also in XML or JSON, providing the interoperability for importing, exporting, and managing the files.

Explaining the process of developing 2D & 3D BIM

As mentioned before, the process of developing BIM is a bottom-up approach. That means the progress is advanced from ideas, design, modeling, and converging to real world objects as the project advances.

BIM transforms the architecture, engineering, and construction industry by offering an integrated, digital representation of a building's physical and functional characteristics. The processes for developing 2D and 3D BIM are different but can be improved through the advances of design.

The process starts with 2D BIM and its utilization to conceptualize and document designs. First, you will produce a sketch or diagram, none of which will be complete design drawings using CAD software, to illustrate the scope and extent of the project. Second, in the schematic design

phase, you will develop and produce a set of project drawings containing detailed floor plans, elevations, and sections, crucial for an understanding of the building, location, and aesthetic. Next, the drawings are enhanced, and the drawings are issued for design development with specifications and notes to finalize the set of drawings needed as construction documents needed to construct the building.

The process switched to 3D BIM and incorporates a complex digital BIM model developed from the 2D model in-to-3D BIM model using software like Autodesk Revit to develop a detailed digital model that represents the building geometry and spatial relationships before being used for structural, mechanical, and other disciplines to develop the model, while executing design coordination to understand and resolve spatial clashes and develop the models collaboratively in one design. The "live" 3D model developed is then used for performance modelling, construction coordination, developed for facility management, building longevity, and optimization of the overall building performance.

In our BIM model, we design the 2D model first so that we can get a 2D floor plan. After that, we set the height of the rooms and add walls. Finally, we design the details of the rooms. This is our process from 2D to 3D.

Difficulties we faced while generating BIM

- not all required architectural or furniture elements are available in the native libraries (Special elements like slide door need to be added from online sources)
- need to calculate a relatively suitable position for some elements like windows, doors and furniture due to the inaccuracy of the original floor plan
- Different interpretations of the reference drawing, for example some think that the balcony is curtain walls, while others think that it is made of fencing

Project Relations with BIM

Collaboration

- enables collaboration between the 7 students on the project
- shared Revit model is the common platform for us to work together on
- Revit provides an easy way to connect the building models together, ensuring the overall model remains cohesive

Data Integration

- Revit stores information about the building components and their properties (e.g. dimensions, thickness, material)
- This data can also be used for future analysis, cost estimation, energy modeling

Visualization

- The 3D building model creates a highly detailed visualization of the final design
- Our group can discuss the project concepts and identify any disputes about the design more effectively
- enable stakeholder engagement (e.g. for client, architect, engineer) and better-informed decision making

Conclusion

In this project, our group has demonstrated collaboration through the development of 3D BIM models. We have divided up the tasks and assigned different workflow for individual members, while resolving some disputed opinions or interpretations of the BIM model. This has led us to more engagement in discussions on the project, and eventually unifying the types of the features and combining them into one BIM model.

Throughout this project, we have broadened our horizons on the skills in operating BIM software as well as other insights, for example, how BIM can be applied to the real world scenarios, which will be invaluable in our future careers. We have also gained hands-on experience on developing BIM with a bottom-up approach, from 2D floor plan to 3D BIM model, while ensuring the dimensions are accurate and applying different techniques to view the BIM model, such as the solar study view, camera view, and walkthrough view.

Despite facing challenges such as time constraints and some of the rooms lacking the furniture for a higher level of detail, we effectively demonstrated our collaborative teamwork and software skills. Overall, this project focuses on effective communication and teamwork in achieving project goals. We have learned from this experience in good BIM practices and the nature of the AEC industry.

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Appendix

Link to the videos for the walkthrough view:

Walkthrough view videos attached: [ExternalWalkthrough.mp4](#) and [InternalWalkthrough.mp4](#)